Interventional Mean Embeddings

for Distributional Causal Inference

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Why distributional causal inference?

Outcomes can be images, protein sequences, text or graphs.

Pixelwise averaging can hide an image's spatial structure.



Different image patterns can have the same mean.

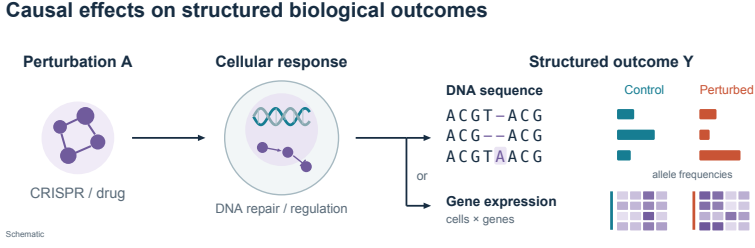
Semi-synthetic OCTMNIST illustration; no rejection in this run.

Zenati and Gretton, NeurIPS 2026.

Why distributional causal inference?

Outcomes can be images, protein sequences, text or graphs.

An average sequence or graph has no canonical meaning without first choosing a representation.



Which repaired DNA sequences occur, and how often?

In Repair-seq, perturbing DNA-repair genes changes the frequencies of repaired sequences after a fixed DNA lesion.

Hussmann et al., Cell, 2021.

The distribution is the causal target

An intervention can change a distribution's modes, their relative mass, or their variability.

The interventional distribution is a natural causal target on the original outcome space.

Muandet et al., Counterfactual Mean Embeddings, 2021.

Our population counterfactual question uses this interventional object.[1]

- Testing: does the intervention change this distribution?
- Sampling: can we generate representative outcomes from this adjusted population?

Scalar distribution functions require an ordering; images and DNA sequences have no natural total order.

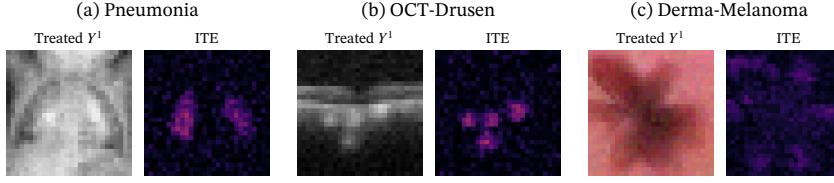
Kernel mean embeddings use outcome similarity instead.

[1] Pearl's ladder: population interventions are level 2; level-3 counterfactuals condition on factual evidence and need additional structure.

Distributional tests on structured outcomes

Compare interventions q, q' through their outcome laws $P^q, P^{q'}$.

$$H_0 : P^q = P^{q'} \qquad \text{against} \qquad H_1 : P^q \neq P^{q'}.$$



Pattern frequency and severity can change with no mean effect.

Can we detect this change from confounded observations?

Learn deep features that make distributional differences easier to detect.

Semi-synthetic MedMNIST.

Leman et al., submitted to ICLR 2027.

Sampling from an adjusted population

Confounding: treatment assignment and outcomes share causes. The treated group can therefore differ from the target population.

Generate outcomes as if the intervention were applied to the whole population.



CelebA: correct the observed hair-colour imbalance.

Sample the intervention's outcomes after adjusting for confounders.

Population-standardisation illustration.

Bozkurt et al., submitted to AISTATS 2027.

Papers and collaborators

Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings
Houssam Zenati, Bariscan Bozkurt and Arthur Gretton
NeurIPS 2025

Semiparametric Efficient Test for Interpretable Distributional Treatment Effects
Houssam Zenati and Arthur Gretton
NeurIPS 2026

Sampling Interventional Distributions with Orthogonal MMD Flows
Bariscan Bozkurt, Hugh Dance, Zonghao Chen, Alexandre Galashov,
Arthur Gretton and Houssam Zenati
2026. Submitted to AISTATS 2027

Doubly Robust Deep Kernel Tests for Structured Outcomes
Robin Leman, Bariscan Bozkurt, Sofiane Tanji, Arthur Gretton and Houssam Zenati
2026. Submitted to ICLR 2027

Collaborators

With thanks to my main collaborators



Bariscan Bozkurt



Arthur Gretton



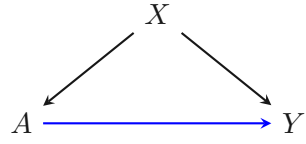
Robin Leman

and also Hugh Dance, Zonghao (Hudson) Chen, Alexandre Galashov and Sofiane Tanji.

The causal graph

At population level, let $O = (X, A, Y) \sim P_0$.

X: pre-treatment confounders; A: treatment; Y: outcome. P_0: observational joint distribution; P_X: its marginal law of X.



The common causes X affect both treatment and outcome.

An intervention changes how treatment is assigned, while retaining the population distribution of X.

We must account for these common causes to identify its outcome law.

The interventional outcome law

Hard intervention: set $A = a$. Its outcome law is $P^a = \mathcal{L}(Y(a))$.

Soft intervention: replace the treatment mechanism $e_0(da \mid x) = P_0(A \in da \mid X = x)$ by a specified $q(da \mid x)$.

$e_0(a \mid x)$: propensity mass/density; $Y(a)$: outcome under a; $\mathcal{L}$: probability law.

Identification from the observational distribution

$$P^q(dy) = \int\int P_0(dy \mid a, x)\, q(da \mid x)\, P_X(dx).$$

- Consistency: $Y = Y(A)$.
- No unmeasured confounding: $Y(a) \perp A \mid X$.
- Overlap: $q(\cdot \mid x) \ll e_0(\cdot \mid x)$.

Overlap means assigning only supported treatments. For discrete A, setting $q = \delta_a$ recovers P^a; δ_a: point mass at a.

Kernels compare structured outcomes

A positive-definite kernel $k(y, v)$ measures similarity between an outcome y and a reference outcome v.

- Images: compare spatial patterns.
- DNA, protein sequences and text: compare subsequences or alignments.
- Graphs: compare common substructures.

The feature map is a similarity profile in the reproducing kernel Hilbert space (RKHS) $\mathcal{H}$:

$$\phi(y) = k(y, \cdot) \in \mathcal{H}.$$

$\langle f, g \rangle_{\mathcal{H}}$: RKHS inner product; $\|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}}$.

Images: Lazebnik et al., 2006; strings and graphs: Haussler, 1999; biological sequences: Saigo, Vert et al., 2004.

Kernel mean embeddings and the witness

For an outcome law P, such as the observed outcome distribution, define its kernel mean embedding (KME):

$$\mu_P = \mathbb{E}_{Y\sim P}[\phi(Y)], \qquad \mu_P(v) = \mathbb{E}_{Y\sim P}[k(Y, v)].$$

The KME records average similarity to every reference outcome v.

To compare two laws P, P', define their witness function:

$$w(v) = \mu_P(v) - \mu_{P'}(v).$$

A positive value means greater average similarity to v under P.

Bounded measurable kernel; existence conditions in backup.

Smola et al., 2007; Gretton et al., 2012.

Interventional mean embeddings

Define the interventional mean embedding (IME):

$$\mu_q = \mathbb{E}_{Y \sim P^q}[\phi(Y)], \qquad \mu_q(v) = \mathbb{E}_{Y \sim P^q}[k(Y, v)].$$

The averaging law is now interventional, rather than observed.

Its interventional witness compares two interventions:

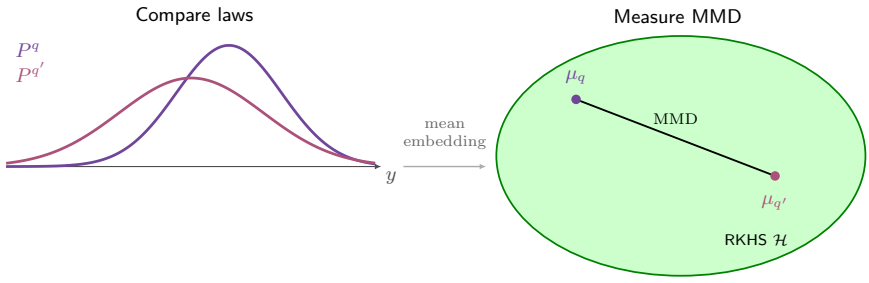
$$w(v) = \mu_q(v) - \mu_{q'}(v).$$

With a characteristic kernel, the embedding identifies the law:

$$\mu_q = \mu_{q'} \quad \Longleftrightarrow \quad P^q = P^{q'}.$$

Muandet et al., Counterfactual Mean Embeddings, 2021.

Testing becomes an embedding comparison



$$\mathrm{MMD}(P^q, P^{q'}) = \|\mu_q - \mu_{q'}\|_{\mathcal{H}}.$$

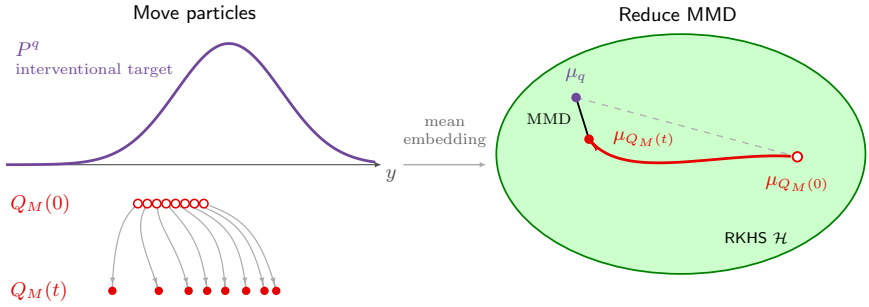
MMD: maximum mean discrepancy. Equal laws give zero distance.

We can also use this embedding distance to generate outcomes.

Characteristic kernel; schematic outcome densities.

Gretton et al., 2012.

Sampling with MMD gradient flow



Move the particles so their embedding approaches the target IME.

Each step follows the gradient of the squared MMD with respect to the particle positions.

$Q_M(t)$: empirical law of M moving particles at time t; $\mu_{Q_M(t)}$: their mean embedding. Smooth kernel; projected RKHS trajectory.

Arbel et al., NeurIPS 2019.

IME estimation

Why adjustment introduces nuisances

From n independent observations $O_i \sim P_0$:

Observed KME: directly average the features.

$$\widehat{\mu}_{P_Y} = \frac{1}{n}\sum_i \phi(Y_i), \qquad \|\widehat{\mu}_{P_Y} - \mu_{P_Y}\|_{\mathcal{H}} = O_p(n^{-1/2}).$$

IME: first adjust for the covariates.

$$\mu_q = \mathbb{E}_X\left[\int m_0(a,X)q(da \mid X)\right] = \mathbb{E}_0\left[\frac{q(A \mid X)}{e_0(A \mid X)}\phi(Y)\right].$$

The CME $m_0(a,x) = \mathbb{E}_0[\phi(Y) \mid A = a, X = x]$ and propensity e_0 are nuisances: auxiliary functions needed for our target.

Estimating them can introduce bias and slow the rate.

P_Y: observed outcome law; $\mathbb{E}_0 = \mathbb{E}_{P_0}$; $\mathbb{E}_X = \mathbb{E}_{P_X}$. Bounded kernel; overlap; $O_p(n^{-1/2})$: root-n stochastic error.

Estimating an IME by prediction

The outcome nuisance is the conditional mean embedding:

$$m_0(a, x) = \mathbb{E}_0[\phi(Y) \mid A = a, X = x].$$

Average a candidate model m over the intervention:

$$m_q(x) = \int m(a, x) q(da \mid x), \qquad \mu_q = \mathbb{E}_X[m_{0,q}(X)].$$

A fitted nuisance $\widehat{m}$ gives the plug-in estimator

$$\widehat{\mu}_q^{\mathrm{PI}} = \frac{1}{n} \sum_i \widehat{m}_q(X_i).$$

Nonparametric CME estimation can be slow; the plug-in estimator generally inherits its error.

Subscript 0: truth; hats: fitted quantities.

Zenati et al., NeurIPS 2025.

Estimating an IME by reweighting

The treatment propensity e_0 gives the weight nuisance r_0:

$$r_0(A, X) = \frac{q(A \mid X)}{e_0(A \mid X)}, \qquad \mu_q = \mathbb{E}_0[r_0(A, X)\phi(Y)].$$

For a discrete treatment set to a,

$$r_0(A, X) = \frac{\mathbf{1}\{A = a\}}{e_0(a \mid X)}.$$

A fitted propensity gives the nuisance $\widehat{r} = q/\widehat{e}$

$$\widehat{\mu}_q^{\mathrm{IPW}} = \frac{1}{n} \sum_i \widehat{r}(A_i, X_i)\phi(Y_i).$$

Can these weights correct first-order errors in the outcome prediction?

Use conditional treatment masses or densities; overlap is required.

Zenati et al., NeurIPS 2025.

Prediction plus a residual correction

Combine the two nuisance models: $\eta = (m, r)$.

m: conditional mean embedding model; r: intervention weight model; $\eta_0 = (m_0, r_0)$: truth.

The feature residual is

$$\phi(Y) - m(A, X).$$

The corrected profile $\xi_q(O; \eta) \in \mathcal{H}$ contributed by one observation is

$$\xi_q(O; \eta) = \underbrace{m_q(X)}_{\text{prediction}} + \underbrace{r(A, X)}_{\text{assignment adjustment}} \underbrace{\{\phi(Y) - m(A, X)\}}_{\text{residual}}.$$

$$\widehat{\mu}_q^{\mathrm{DR}} = \frac{1}{n} \sum_i \widehat{\xi}_q(O_i).$$

$\widehat{\xi}_q(O_i) = \xi_q(O_i; \widehat{\eta}_{-i})$: use models fitted outside observation i's fold.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Why is it doubly robust?

$$\xi_q(O;\eta) = m_q(X) + r(A,X)\{\phi(Y) - m(A,X)\}.$$

Correct outcome model: residuals average to zero:

$$\mathbb{E}_0[\phi(Y) - m_0(A,X) \mid A, X] = 0.$$

The prediction already targets μ_q.

Correct weights: the correction cancels prediction error:

$$\mathbb{E}_0[r_0 m(A,X)] = \mathbb{E}_X[m_q(X)].$$

Hence $\mathbb{E}_0\xi_q(O;(m,r_0)) = \mathbb{E}_0[r_0\phi(Y)] = \mu_q$.

Either correct model removes the population bias.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

When both models are approximate

Average the corrected profile at the population level:

$$h_\eta = \mathbb{E}_0[\xi_q(O;\eta)], \qquad h_{\eta_0} = \mu_q, \quad \eta_0 = (m_0, r_0).$$

The two nuisance errors multiply:

$$h_\eta - \mu_q = -\mathbb{E}_0[(r - r_0)(m - m_0)], \qquad \|h_\eta - \mu_q\|_{\mathcal{H}} \le \rho_r \rho_m.$$

Weight error: $\rho_r = \|r - r_0\|_{L^2}$.

CME error: $\rho_m = \|m - m_0\|_{L^2(\mathcal{H})}$.

The correction removes each nuisance's first-order bias.

$\|f\|_{L^2}^2 = \mathbb{E}_0|f(A,X)|^2; \|g\|_{L^2(\mathcal{H})}^2 = \mathbb{E}_0\|g(A,X)\|_{\mathcal{H}}^2.$

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Recovering the root-n rate

For the fitted nuisance errors ρ_r, ρ_m, under regularity:

$$\|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}} = O_p\left(\underbrace{n^{-1/2}}_{\text{sampling noise}} + \underbrace{\rho_r \rho_m}_{\text{nuisance bias}}\right).$$

The nuisances can converge slowly, provided their product is small.

If $\rho_r \rho_m = o_p(n^{-1/2})$, nuisance bias is negligible relative to sampling noise.

We recover the root-n scale of the ordinary empirical KME.

Cross-fitting, controlled score second moments and consistency. $O_p(a_n)$: order a_n in probability; $o_p(a_n)$: negligible relative to a_n.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

What the applications inherit

Compare the two interventional embeddings directly. Here $\widehat{\mu}_q$ and $\widehat{\mu}_{q'}$ are their DR estimates.

Comparisons inherit estimation accuracy

$$\begin{aligned}&\left|\|\widehat{\mu}_q - \widehat{\mu}_{q'}\|_{\mathcal{H}} - \|\mu_q - \mu_{q'}\|_{\mathcal{H}}\right| \\ &\qquad \le \|\widehat{\mu}_q - \mu_q\|_{\mathcal{H}} + \|\widehat{\mu}_{q'} - \mu_{q'}\|_{\mathcal{H}}.\end{aligned}$$

- Finite-location testing: witness estimates attain the semiparametric covariance bound at fixed prototypes.
- Sampling: MMD particle velocities depend linearly on the IME; their nuisance bias is second order.

We first use these properties to compare two interventional laws.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Global testing

Global distributional testing

$$H_0 : P^q = P^{q'}, \qquad H_1 : P^q \neq P^{q'}.$$

Global: compare expectations in all RKHS directions. With a characteristic kernel,

$$H_0 \iff \mu_q = \mu_{q'} \iff \mathrm{MMD}^2 = 0.$$

Subtract the two fitted corrected profiles for each observation:

$$\widehat{d}_i = \widehat{\xi}_q(O_i) - \widehat{\xi}_{q'}(O_i).$$

$\widehat{d}_i$: corrected profile difference; its average estimates $\mu_q - \mu_{q'}$.

Squared MMD has zero first-order derivative at the null.

Its null fluctuation is quadratic: degeneracy explained in backup.

Global testing can also use deep features learned for test power.

Zenati et al., NeurIPS 2025.
Leman et al., submitted to ICLR 2027.

Gaussian calibration through cross-MMD

1. Fit. Fit nuisances separately. On evaluation splits I_1, I_2, compute $\widehat{d}_i = \widehat{\xi}_q(O_i) - \widehat{\xi}_{q'}(O_i)$.

2. Project. Use I_2 to choose a direction; project I_1 onto it:

$$\bar{d}_2 = \frac{1}{|I_2|} \sum_{j \in I_2} \widehat{d}_j, \qquad f_i = \langle \widehat{d}_i, \bar{d}_2 \rangle_{\mathcal{H}} \quad (i \in I_1).$$

3. Test. With $\bar{f}, s_f$ the mean and standard deviation on I_1,

$$T = \frac{\sqrt{|I_1|}\,\bar{f}}{s_f} \xrightarrow[H_0]{d} \mathcal{N}(0,1), \qquad \text{reject if } T > z_{1-\alpha}.$$

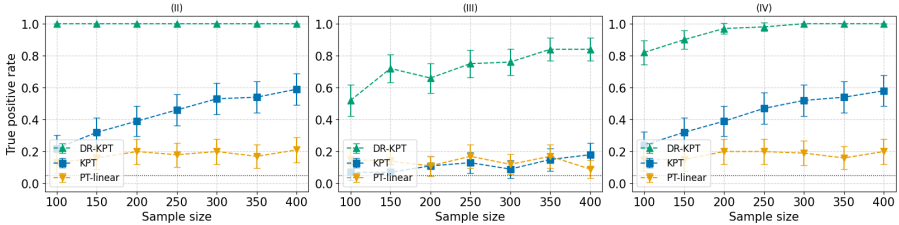
DR makes nuisance bias negligible at the testing scale.

A plug-in bias can shift the null; splitting and studentisation give the Gaussian calibration.

Mean-square consistent scores; product bias $o_p(n^{-1/2})$ for both targets; finite fourth moments and nondegenerate null covariance. $z_{1-\alpha}$: normal quantile; $\xrightarrow{d}$: distributional limit.

Zenati et al., NeurIPS 2025; Kim and Ramdas, Bernoulli 2024.

Effects beyond a mean shift



A mean-preserving mixture still changes the outcome law.

A semi-synthetic image-distribution comparison (dSprites):

RBF method	Null rejection	Power
Reweighted baseline	0.401	0.054
Doubly robust test	0.024	0.656

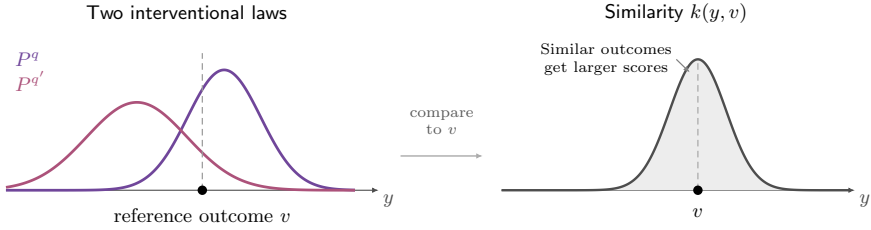
Zenati et al., NeurIPS 2025.

Interpretable testing

Which outcomes reveal the change?

Global testing asks whether the laws differ. We now ask which outcomes help reveal the difference.

Choose a reference outcome v.



Kernel similarity $k(y, v)$ is large when y resembles v.

For images, v is a reference image. One-dimensional schematic shown.

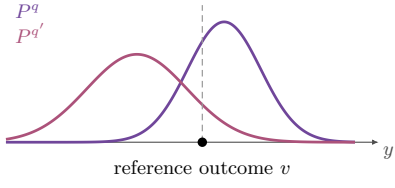
Zenati and Gretton, NeurIPS 2026.

Compare average similarity

Average $k(Y, v)$ under each intervention:

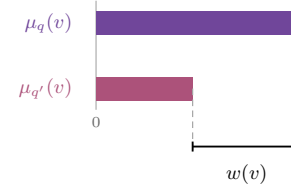
$$\mu_q(v) = \mathbb{E}_{Y \sim P^q}[k(Y, v)], \qquad \mu_{q'}(v) = \mathbb{E}_{Y \sim P^{q'}}[k(Y, v)].$$

Two interventional laws



average similarity

Mean similarity to the same v



$$w(v) = \mu_q(v) - \mu_{q'}(v).$$

Repeat at a few reference outcomes; test their differences jointly.

These reference outcomes are called test locations, or prototypes.

Zenati and Gretton, NeurIPS 2026.

1. From the witness to a finite signal

Fix a kernel k and J reference outcomes (test locations) $V = (v_1, \dots, v_J)$.
Read the interventional witness at these locations:

$$\mathbf{w}(V) = \big(\mu_q(v_j) - \mu_{q'}(v_j)\big)_{j=1}^{J} = \big(w(v_j)\big)_{j=1}^{J} \in \mathbb{R}^J.$$

We now have a finite-dimensional mean-testing problem.

$$P^q = P^{q'} \quad \Longrightarrow \quad \mathbf{w}(V) = 0.$$

In the ordinary ME test, we average similarities from two samples. Here, we observe $O = (X, A, Y) \sim P_0$, not samples from the two interventional laws.

We need observable features whose mean is this causal signal.

A fixed finite set of locations can miss a distributional difference.

Jitkrittum et al., NeurIPS 2016; Zenati and Gretton, NeurIPS 2026.

2. Compute corrected similarities

With separately fitted nuisances, evaluate prediction plus residual correction at each v_j under both interventions:

$$\widehat{d}_i(v_j) = \widehat{\xi}_q(O_i)(v_j) - \widehat{\xi}_{q'}(O_i)(v_j).$$

At true nuisances, these differences have means $w(v_j)$.

Average over observations and estimate the covariance:

$$\widehat{w}_j = \frac{1}{n}\sum_{i=1}^{n} \widehat{d}_i(v_j), \qquad \widehat{\mathbf{w}} = (\widehat{w}_1, \ldots, \widehat{w}_J)^\top,$$

$$\widehat{\Sigma}_{j\ell} = \frac{1}{n-1}\sum_{i=1}^{n} \big(\widehat{d}_i(v_j) - \widehat{w}_j\big)\big(\widehat{d}_i(v_\ell) - \widehat{w}_\ell\big).$$

The DR correction removes first-order nuisance bias in these means.

$j, \ell = 1, \ldots, J$; n: size of the current evaluation sample. Write $\widehat{\mathbf{w}}(V), \widehat{\Sigma}(V)$ when varying the locations.

Zenati and Gretton, NeurIPS 2026.

3. Normalize by the efficient covariance

The population covariance of the corrected similarity vector is

$$\Sigma(V) = \mathrm{Cov}_0\Big[\big(\xi_q(O)(v_j) - \xi_{q'}(O)(v_j)\big)_{j=1}^{J}\Big].$$

Define one covariance-normalised discrepancy:

$$D(V) = \mathbf{w}(V)^\top \Sigma(V)^{-1} \mathbf{w}(V).$$

For one location: squared mean difference divided by its variance. With several: account for noise *and* correlations.

Why this covariance? The centred corrected vector is the canonical gradient of $\mathbf{w}(V)$.

$\Sigma(V)$ is its semiparametric efficiency bound.

$\xi_q(O) = \xi_q(O; \eta_0)$ uses true nuisances; similarly for q'. Cov_0: covariance under P_0; assume $\Sigma(V)$ positive definite.

Zenati and Gretton, NeurIPS 2026.

4. Learn locations with this discrepancy

Fit nuisances $\longrightarrow$ learn locations $\longrightarrow$ test. Use three disjoint samples; fix the kernel and J.

On the location-training sample, compute $\widehat{\mathbf{w}}(V), \widehat{\Sigma}(V)$ with the preceding formulas. Then approximately maximise the stabilised estimate of $D(V)$:

$$\widehat{V} \approx \arg\max_{V} \widehat{\mathbf{w}}(V)^{\top} \big(\widehat{\Sigma}(V) + \tau I_J\big)^{-1} \widehat{\mathbf{w}}(V).$$

Why this objective? For small, root-n alternatives, $nD(V)$ governs limiting test power.

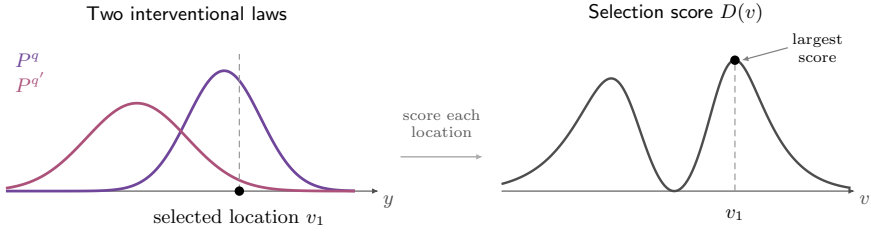
Next: choose one informative location in the same 1D example.

$\tau > 0$: training ridge; I_J: identity. The ridge gives a stabilised power proxy. Local-power explanation in backup.

Jitkrittum et al., NeurIPS 2016; Zenati and Gretton, NeurIPS 2026.

Choose an informative location

For one location, compare $D(v) = w(v)^2/\Sigma(v)$ as v moves along the outcome axis.



Choose v_1 with the strongest difference relative to its variance.

$\Sigma(v)$: variance of the corrected similarity difference. Population illustration; in practice, learn the estimated score.

Zenati and Gretton, NeurIPS 2026.

5. Freeze the locations, then test

Freeze the nuisance fits and $\widehat{V}$. On n untouched observations, recompute $\widehat{\mathbf{w}}$ and $\widehat{\Sigma}$ at these locations.

Calibrate. Under the null, the DR first-order limit gives

$$\sqrt{n}\,\widehat{\mathbf{w}} \Rightarrow \mathcal{N}(0, \Sigma).$$

Whitening this mean gives the Hotelling statistic:

$$T_n = n\widehat{\mathbf{w}}^\top (\widehat{\Sigma} + \gamma_n I_J)^{-1} \widehat{\mathbf{w}} \Rightarrow \chi^2_J.$$

Reject the null of equal interventional laws when

$$T_n > \chi^2_{J,1-\alpha}.$$

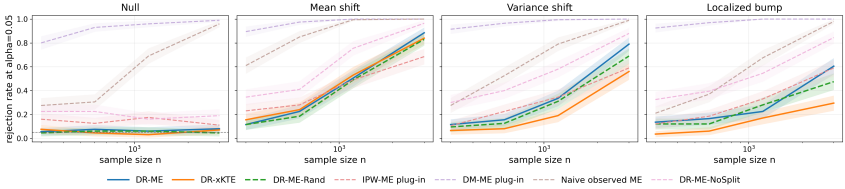
One Hotelling test combines all selected similarities.

α: significance level; $\chi^2_{J,1-\alpha}$: chi-square quantile. $\gamma_n \to 0$: test ridge, distinct from training τ.

Conditional on fitting and learning: fixed J; $\widehat{\Sigma}, \Sigma(\widehat{V}) \to \Sigma$, a positive-definite limit; mean-square score consistency, product bias $o_p(n^{-1/2})$ and moment control.

Zenati and Gretton, NeurIPS 2026.

Power from informative locations



Learned locations improve detection of localised changes.

At outcome dimension 25, the whitening ablation reports:

Location criterion	Power
Full covariance whitening	0.760
Raw witness	0.380
Random locations	0.295

Synthetic benchmarks, level 0.05; plot and table use separate protocols.

Zenati and Gretton, NeurIPS 2026.

Sampling

Generating outcomes under intervention

Testing compares two interventional laws. Sampling asks what outcomes a specified law produces.

- Existing approaches fit a generative model using a causally adjusted objective.
- This can require expensive optimisation for each new target, in addition to fitting the nuisance models.

Our approach: optimise a collection of samples directly.

Estimate the IME once, then move the samples to match it.

Choose a loss $\longrightarrow$ correct its target $\longrightarrow$ move particles.

Bozkurt et al., submitted to AISTATS 2027.

1. Sampling as distributional optimisation

Let Q be the distribution of the outcomes we generate. Choose it to minimise

$$F(Q) = \tfrac{1}{2}\|\mu_Q - \mu_q\|^2_{\mathcal{H}}.$$

A characteristic kernel makes the population minimiser P^q.

Expand the loss to see what we need from the target:

$$F(Q) = \underbrace{\tfrac{1}{2}\mathbb{E}_{Q\otimes Q}k(Z,Z')}_{\text{our candidate outcomes}} - \underbrace{\mathbb{E}_Q[\mu_q(Z)]}_{\text{target similarity profile}} + \text{constant}.$$

One IME supplies the objective for every candidate distribution.

Z, Z': independent draws from Q; $\mu_q(z) = \mathbb{E}_{Y\sim P^q}k(Y,z)$. The omitted constant does not depend on Q.

Bozkurt et al., submitted to AISTATS 2027.

2. Correct the target once

Prediction alone can introduce first-order bias into the target. Recall the prediction plus weighted residual estimator:

$$\widehat{\mu}_q^{\mathrm{DR}} = \frac{1}{n}\sum_i \left[\widehat{m}_q(X_i) + \widehat{r}(A_i, X_i)\{\phi(Y_i) - \widehat{m}(A_i, X_i)\}\right].$$

m: CME; r: intervention weight; $m_q(x) = \int m(a,x)q(da \mid x)$. $\eta = (m, r)$; subscript 0: truth; ξ_q: corrected profile in brackets.

Correct weights average residuals under the intervention, cancelling prediction bias. With both models approximate,

$$\mathbb{E}_0 \xi_q(O;\eta) - \mu_q = -\mathbb{E}_0[(r - r_0)(m - m_0)].$$

Only the product of their errors remains in the population bias.

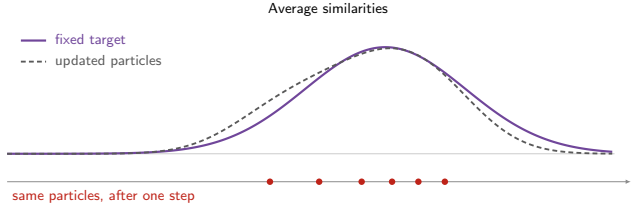
Substitute $\widehat{\mu}_q^{\mathrm{DR}}$ in the preceding loss. The same correction protects every candidate Q.

Bozkurt et al., submitted to AISTATS 2027.

3. Moving the particles

Move M particles $z_1, \ldots, z_M$ to match the fixed target.

Recompute with the moved particles; keep the target fixed.



$$f_t(z) = \mu_{Q_M(t)}(z) - \widehat{\mu}_q^{\mathrm{DR}}(z), \qquad \frac{dz_j}{dt} = -\nabla_z f_t(z_j).$$

$Q_M(t) = M^{-1} \sum_j \delta_{z_j(t)}$: current particle distribution; δ_z: point mass. t: flow time; ∇_z: spatial gradient. 1D illustration, smooth Gaussian kernel.

Bozkurt et al., submitted to AISTATS 2027.

4. The particle update in practice

Fit and freeze. Fit the nuisances; store the target profile

$$\widehat{\mu}_q^{\mathrm{DR}}(z) = \sum_{i=1}^{n} c_i k(z, \bar{z}_i), \qquad \bar{z}_i = E(Y_i).$$

E: fixed encoder; here k, μ_q refer to encoded outcomes. c_i: signed DR coefficients (formula in backup), not resampling probabilities.

Move. Initialise M particles at observed encodings. At iteration r, update all positions simultaneously:

$$z_j^{(r+1)} = z_j^{(r)} + \epsilon_r \left[\underbrace{\sum_{i=1}^{n} c_i \nabla_1 k(z_j^{(r)}, \bar{z}_i)}_{\text{fixed target force}} - \underbrace{\frac{1}{M} \sum_{\ell=1}^{M} \nabla_1 k(z_j^{(r)}, z_\ell^{(r)})}_{\text{particle interaction}} \right].$$

Decode. After R updates, return $D(z_j^{(R)})$.

D: fixed decoder; $E = D = \mathrm{Id}$ gives outcome-space sampling. $\epsilon_r > 0$: step size; ∇_1: first-argument gradient. Freeze $c_i, \bar{z}_i$.

Bozkurt et al., submitted to AISTATS 2027.

5. What the DR target protects

At the same current particle cloud, only the target force changes when μ_q is replaced by $\widehat{\mu}_q^{\mathrm{DR}}$:

$$\sup_z \|\nabla_z(\widehat{\mu}_q^{\mathrm{DR}} - \mu_q)(z)\|_2 \le L_\phi \|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}}.$$

Under the earlier estimation conditions, embedding error has order

$$\underbrace{n^{-1/2}}_{\text{sampling noise}} + \underbrace{\rho_r \rho_m}_{\text{nuisance bias}} .$$

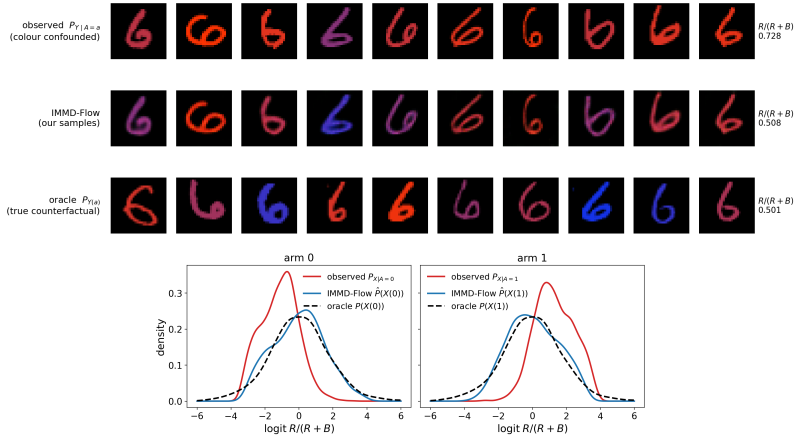
First-order nuisance bias is removed from the force at every cloud.

Final sampling accuracy also depends on optimisation progress and the number of particles.

ρ_m, ρ_r: CME and weight errors. L_ϕ bounds the feature derivative; smooth kernel. Formal orthogonality in backup.

Bozkurt et al., submitted to AISTATS 2027.

Correcting the whole colour distribution



Digit identity follows the intervention. Colour follows the population.

Bozkurt et al., submitted to AISTATS 2027.

Computational cost of interventional sampling

Nuisance models estimate the target IME. Particles then optimise directly against that fixed target.

The saving comes from avoiding an additional generator fit.

Fitting + generation (seconds)	DR–IMMD	DecFM plug-in
ColorMNIST	38.4	423.2
CelebA	700.5	8849.5

11–42× lower total cost across the reported DecFM variants.

Totals exclude shared preprocessing, decoding and evaluation. DecFM generation alone is faster.

Bozkurt et al., submitted to AISTATS 2027.

One IME, several causal procedures

A full interventional distribution becomes a computable object.

- DR estimation: an orthogonal score with product-order bias.
- Global testing: calibrated evidence of a distributional change.
- Interpretable testing: efficient finite-signal inference and learned locations.
- Sampling: an orthogonal objective and robust transport forces.

The estimation properties become properties of the applications.

Zenati et al., NeurIPS 2025; Zenati and Gretton, NeurIPS 2026; Bozkurt et al., submitted to AISTATS 2027.

Backup slides

Identification and IME estimation

Confounding · Well-defined score · Orthogonality

Valid probability paths · Asymptotic linearity and efficiency

Distributional testing

MMD degeneracy · Complementary locations

Local power geometry · Regular local alternatives

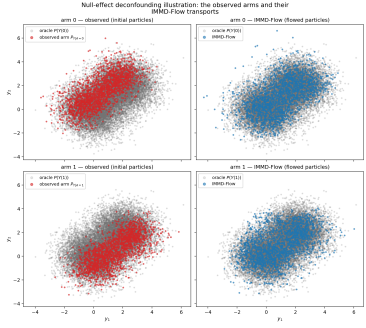
Sampling

Orthogonal optimisation · Kernel regression

DR coefficients · Generated-law guarantee

Causal target: confounding

Observed groups can differ even when the intervention has no effect.



Here $\mathcal{L}_0(Y \mid A = 0) \neq \mathcal{L}_0(Y \mid A = 1)$, but $P^0 = P^1$.

Rows: treatment arms. Left: observed samples (red). Right: generated samples (blue). Grey: the common interventional target.

Bozkurt et al., submitted to AISTATS 2027.

Backup contents

Return to talk

Estimation: well-defined profiles

For $\eta = (m, r)$, the corrected profile is

$$\xi_q(O; \eta) = m_q(X) + r(A, X)\{\phi(Y) - m(A, X)\}.$$

$m_q(x) = \int m(a, x) q(da \mid x)$ averages the candidate CME over the intervention.

Suppose $k(y, y) \le \kappa^2 < \infty$, $r_0, r \in L^2$ and $m \in L^2(\mathcal{H})$. Cauchy–Schwarz then gives

$$\mathbb{E}_0 \|\xi_q(O; \eta)\|_{\mathcal{H}} < \infty.$$

Its population mean exists in $\mathcal{H}$.

This ensures integrability; inference also needs finite variance.

$O = (X, A, Y) \sim P_0$; $\phi(y) = k(y, \cdot)$; e_0: treatment propensity; $r_0 = dq/de_0$ with $q \ll e_0$. $\|r\|_{L^2}^2 = \mathbb{E}_0 |r(A, X)|^2$ and $\|m\|_{L^2(\mathcal{H})}^2 = \mathbb{E}_0 \|m(A, X)\|_{\mathcal{H}}^2$. Standard Borel data spaces, separable $\mathcal{H}$, strongly measurable ϕ.

Zenati, Bozkurt and Gretton, NeurIPS 2025; Bozkurt et al., submitted to AISTATS 2027.

Estimation: Neyman orthogonality

Keep P_0 and q fixed. Perturb the two nuisance functions:

$$\eta_t = (m_0 + t\dot{m},\ r_0 + t\dot{r}), \qquad \eta_0 = (m_0, r_0).$$

Here t is a small real step; $\dot{m}, \dot{r}$ are fixed directions.

The population product-error identity becomes

$$\mathbb{E}_0 \xi_q(O; \eta_t) - \mu_q = -t^2 \mathbb{E}_0[\dot{r}(A, X)\dot{m}(A, X)].$$

$$\left.\frac{d}{dt}\mathbb{E}_0 \xi_q(O; \eta_t)\right|_{t=0} = 0.$$

The mean corrected profile has no first-order nuisance bias.

ξ_q: corrected profile; $\mathbb{E}_0 \xi_q(O; \eta_0) = \mu_q$. $\dot{m} \in L^2(\mathcal{H})$, $\dot{r} \in L^2$ under the observed (A, X) law. Linear paths are valid in this function space; probability-model paths are covered too.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Estimation: valid probability paths

Valid probability paths preserve the same second-order bias.

Treatment assignment: for binary A, perturb its propensity:

$$e_t(1 \mid x) = e_0(1 \mid x) + t\dot{e}(x).$$

$e_0(1 \mid x) \in [\varepsilon, 1-\varepsilon]$, $0 < \varepsilon < 1/2$; bounded direction $|\dot{e}(x)| \le 1$; $|t| < \varepsilon/2$. Set $e_t(0 \mid x) = 1 - e_t(1 \mid x)$.

Outcome model: tilt its conditional law by a bounded function s:

$$P_t(dy \mid a, x) = (1 + t\, s(y, a, x))P_0(dy \mid a, x).$$

$\mathbb{E}_0[s(Y, A, X) \mid A, X] = 0$ and $|t\, s(y, a, x)| < 1$ ensure a probability law.

Nuisance changes are $O(t)$; corrected-score bias is $O(t^2)$.

Keep P_0 fixed when evaluating $\mathbb{E}_0\xi_q$; only the nuisance inputs change.

Bozkurt et al., submitted to AISTATS 2027.

Estimation: asymptotic linearity

For independent $O_i \sim P_0$, with cross-fitting and negligible nuisance bias,

$$\sqrt{n}(\widehat{\mu}_q^{\mathrm{DR}} - \mu_q) = \frac{1}{\sqrt{n}} \sum_i \{\xi_q(O_i; \eta_0) - \mu_q\} + o_p(1).$$

Here $\eta_0 = (m_0, r_0)$ contains the true nuisance functions.

Asymptotic linearity: to first order, we average the true corrected profiles.

Their centred version is the canonical gradient: it gives the smallest asymptotic covariance for regular estimation of fixed embedding coordinates.

Evaluating the IME at fixed locations inherits this efficiency.

Foldwise conditions: finite true-score second moment; fitted scores converge in mean square; $\|\widehat{r} - r_0\|_{L^2} \|\widehat{m} - m_0\|_{L^2(\mathcal{H})} = o_p(n^{-1/2})$. The remainder is $o_p(1)$ in $\mathcal{H}$ norm.

Zenati et al., NeurIPS 2025; Luedtke and Chung, AoS 2024.

Testing: why squared MMD is degenerate

With true nuisances, let $d_i = \xi_q(O_i; \eta_0) - \xi_{q'}(O_i; \eta_0)$. Under equal interventional laws, $\mathbb{E}_0 d_i = 0$.

$$U_n = \frac{1}{n(n-1)} \sum_{i \neq j} \langle d_i, d_j \rangle_{\mathcal{H}}.$$

Degenerate: the conditional mean given one observation is zero.

$$\mathbb{E}_0[\langle d_i, d_j \rangle_{\mathcal{H}} \mid d_i] = \langle d_i, \mathbb{E}_0 d_j \rangle_{\mathcal{H}} = 0.$$

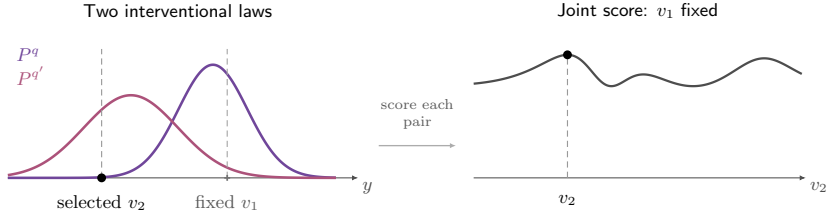
$$nU_n \Rightarrow \sum_{\ell} \lambda_\ell (Z_\ell^2 - 1), \qquad Z_\ell \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

λ_ℓ: eigenvalues of the covariance operator of d_i. Independent oracle scores; separable $\mathcal{H}$; finite second moment and nonzero covariance.

Gretton et al., JMLR 2012; Kim and Ramdas, Bernoulli 2024.

Testing: add a complementary location

Fix the first reference outcome v_1. Choose v_2 by scoring the pair, accounting for noise *and* correlation.



The second comparison adds information beyond the first.

Joint score: $\mathbf{w}(V)^\top (\Sigma(V) + \tau I_2)^{-1} \mathbf{w}(V)$, with $V = (v_1, v_2)$ and $\tau = 10^{-4}$.

$\mathbf{w}(V) = (\mu_q(v_j) - \mu_{q'}(v_j))_{j=1}^2$; $\Sigma(V)$: covariance of the two true corrected similarity differences; I_2: identity. Population illustration.

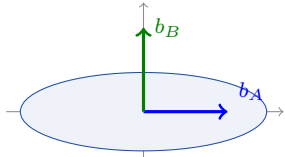
Zenati and Gretton, NeurIPS 2026.

Backup contents

Return to talk

Testing: signal in noise units

Local changes have size $1/\sqrt{n}$, the scale of sampling noise. At fixed test locations V, suppose $\sqrt{n}\,\mathbf{w}_n(V) \to b \in \mathbb{R}^J$.



An equal-λ ellipse

Two shifts of equal raw size. The vertical direction is less noisy.

Whitening measures the shift relative to its covariance:

$$\lambda = b^\top \Sigma^{-1} b.$$

Larger λ gives greater limiting power, at fixed J and level α.

Here b_B is more detectable than b_A.

$\mathbf{w}_n(V) = (\mu_{q,n}(v_j) - \mu_{q',n}(v_j))_{j=1}^J$: true witness values under the local alternative.

$\Sigma = \mathrm{Cov}_0[(d(O)(v_j))_{j=1}^J]$ is positive definite, where $d(O) = \xi_q(O;\eta_0) - \xi_{q'}(O;\eta_0)$ uses true null nuisances. Local-limit assumptions.

Zenati and Gretton, NeurIPS 2026.

Backup contents

Return to talk

Testing: the local-power limit

Fix J test locations V. Take n observations from a regular law path $P_{c/\sqrt{n}}$ starting at a null law P_0; $c \in \mathbb{R}$ is fixed.

The true witness vector satisfies $\sqrt{n}\,\mathbf{w}_n(V) \to b$. The efficient DR estimate then obeys

$$\sqrt{n}\,\widehat{\mathbf{w}} \Rightarrow \mathcal{N}(b, \Sigma).$$

With a consistent covariance estimate and $\gamma_n \to 0$,

$$T_n = n\widehat{\mathbf{w}}^\top (\widehat{\Sigma} + \gamma_n I_J)^{-1} \widehat{\mathbf{w}} \Rightarrow \chi^2_J(\lambda), \qquad \lambda = b^\top \Sigma^{-1} b.$$

Power at level α: $\Pr\{\chi^2_J(\lambda) > \chi^2_{J,1-\alpha}\}$ in the limit.

$\mathbf{w}_n(V) = (\mu_{q,n}(v_j) - \mu_{q',n}(v_j))_{j=1}^J$; $\Sigma = \mathrm{Cov}_0[(d(O)(v_j))_{j=1}^J]$ is positive definite, with $d(O) = \xi_q(O;\eta_0) - \xi_{q'}(O;\eta_0)$.

Regular path: quadratic-mean differentiable within the causal model. Assume the efficient expansion at P_0; I_J: identity; $\chi^2_{J,1-\alpha}$: central chi-square quantile.

Zenati and Gretton, NeurIPS 2026.

Sampling: orthogonality of the forces

For fixed Q and nuisance inputs $\eta = (m, r)$, the witness is

$$f_{Q,\eta}(z) = \mu_Q(z) - \mathbb{E}_0[\xi_q(O;\eta)(z)].$$

For real t, $\eta_t = \eta_0 + t(\dot{m}, \dot{r})$ gives

$$f_{Q,\eta_t}(z) - f_{Q,\eta_0}(z) = t^2 \mathbb{E}_0[\dot{r}\,\dot{m}](z).$$

The witness and transport force have zero nuisance derivative.

This holds at every candidate distribution Q.

P_0, q remain fixed; m: CME; r: weight; η_0: truth. Directions $\dot{m} \in L^2(\mathcal{H})$, $\dot{r} \in L^2$ under the observed (A, X) law; bounded feature derivative transfers the result to $-\nabla_z f$.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: kernel regression in closed form

For n observations, set $S_i = (A_i, X_i)$ and $\bar{z}_i = E(Y_i)$. Use an input kernel k_S on treatment–context pairs.

$$K_{ij} = k_S(S_i, S_j), \qquad \mathbf{k}_s = (k_S(S_1, s), \ldots, k_S(S_n, s))^\top.$$

With ridge $\lambda > 0$, set $B_\lambda = (K + n\lambda I_n)^{-1}$. Kernel regression gives

$$\widehat{m}(s) = \sum_{i=1}^{n} [B_\lambda \mathbf{k}_s]_i \, \phi(\bar{z}_i).$$

Predictions combine the observed outcome features.

Next: combine prediction with the weighted residual.

$s = (a, x)$: a treatment–context pair; I_n: identity; $\phi(z) = k(z, \cdot)$; E: fixed encoder, or the identity.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: the DR coefficients

Same-sample full ridge: define the intervention averages

$$[K_q]_{ij} = \int k_S(S_i, (a, X_j)) q(da \mid X_j), \qquad \widehat{\mathbf{r}} = (\widehat{r}(A_i, X_i))_{i=1}^n.$$

The corrected target is $\widehat{\mu}_q^{\mathrm{DR}} = \sum_i c_i \phi(\bar{z}_i)$, where

$$\mathbf{c} = \frac{1}{n}\left[\underbrace{B_\lambda K_q \mathbf{1}_n}_{\text{prediction}} + \underbrace{(I_n - B_\lambda K)\widehat{\mathbf{r}}}_{\text{residual correction}}\right].$$

Compute these coefficients once; freeze them during generation.

Signed coefficients are not resampling probabilities.

K: input Gram matrix; $B_\lambda = (K + n\lambda I_n)^{-1}$; $\mathbf{1}_n$: all-ones vector; $\mathbf{c} = (c_1, \ldots, c_n)^\top$. Regression definitions.

Bozkurt et al., submitted to AISTATS 2027.

Sampling: three sources of error

For a fixed representation, let $e_n = \|\widehat{\mu}_q^{\mathrm{DR}} - \mu_q\|_{\mathcal{H}}$.

After R updates with M particles, the guarantee separates

$$\begin{aligned} \|\mu_{Q_M^{(R)}} - \mu_q\|_{\mathcal{H}} \lesssim\ & e^{-cR^{1-2\alpha}} && \text{optimisation} \\ & + (1 + R^{\alpha})e_n && \text{target estimation} \\ & + R^{\alpha}\sqrt{\log(R/\delta)/M} && \text{finite particles.} \end{aligned}$$

More iterations reduce the first term but can amplify target error.

DR makes the nuisance bias in e_n second order.

$Q_M^{(R)}$: particle law; $c > 0$: constant; $0 \le \alpha < 1/2$: noise-decay exponent; $0 < \delta < 1$: failure probability; $\lesssim$: up to fixed constants.

Conditional probability at least $1 - \delta$ under smooth noisy updates, gradient dominance and a bounded computed target. Decoding needs further conditions.

Bozkurt et al., submitted to AISTATS 2027.